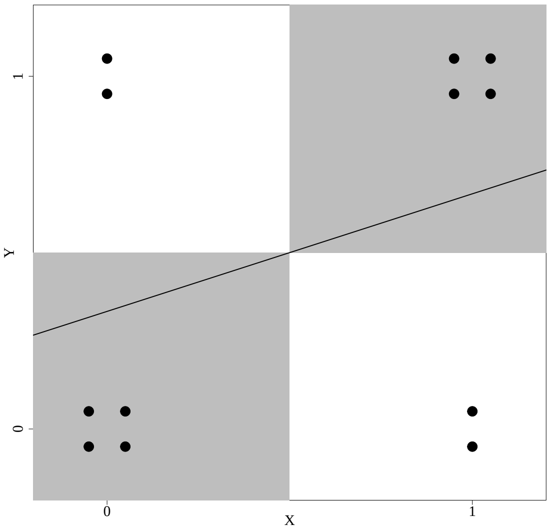
Descriptive Multiple Linear Regression

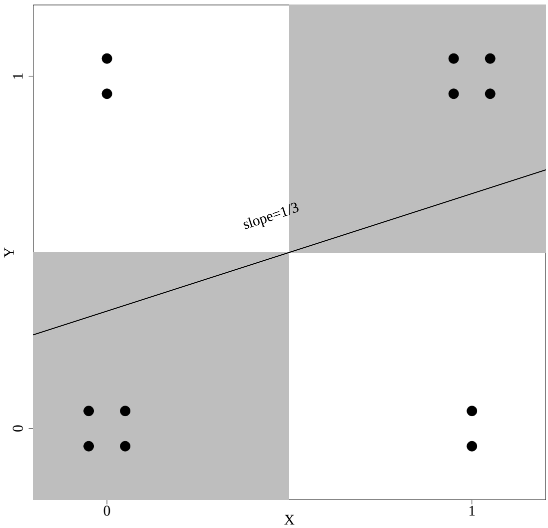
October 19, 2023

Outline

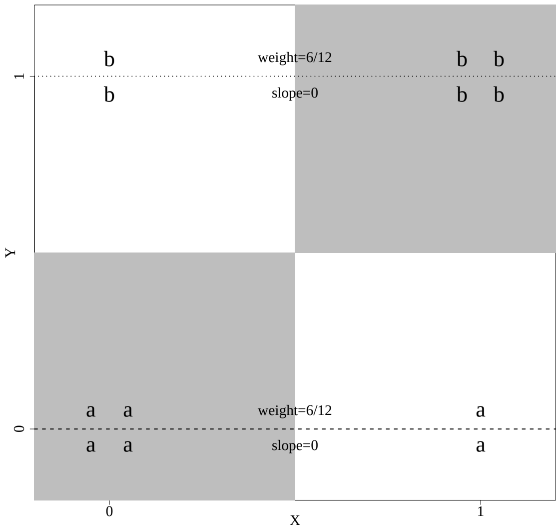
Introductory Exercise



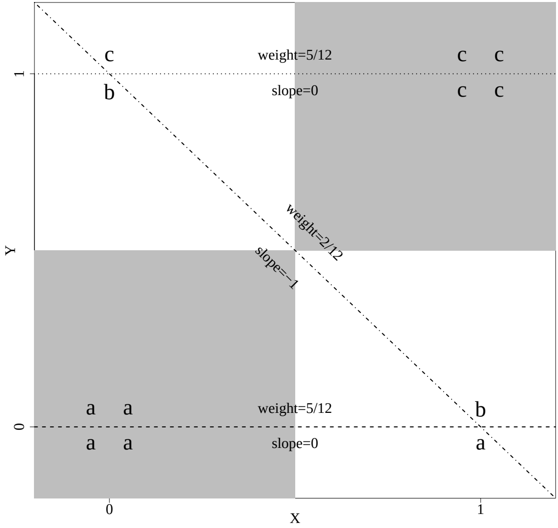
Introductory Exercise



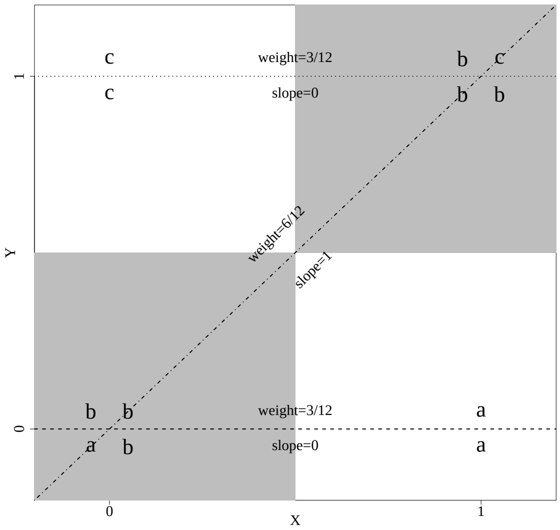
Introductory Exercise



Introductory Exercise



Introductory Exercise



Outline

More Than One Conditioning Variable

- Often we want averages conditional on more than one variable.
- As with continuous conditioning variables, there are many ways to do this.
- Today we will look at two commonly used “linear” versions.
- Unless the two conditioning variables are binary and an interaction is included, these approaches will induce misspecification.

Multiple Regression with 2 Conditioning Variables

- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i$ for $i = 1, ..., N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, and B_2, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i changes with the inclusion of more variables.

Multiple Regression with 2 Conditioning Variables

- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i$ for $i = 1, ..., N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, and B_2, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i changes with the inclusion of more variables.

Multiple Regression with 2 Conditioning Variables

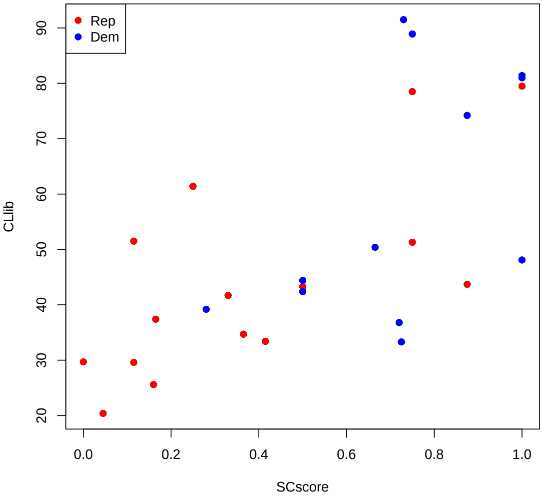
- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i$ for $i = 1, ..., N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, and B_2, to minimize the SSR, where

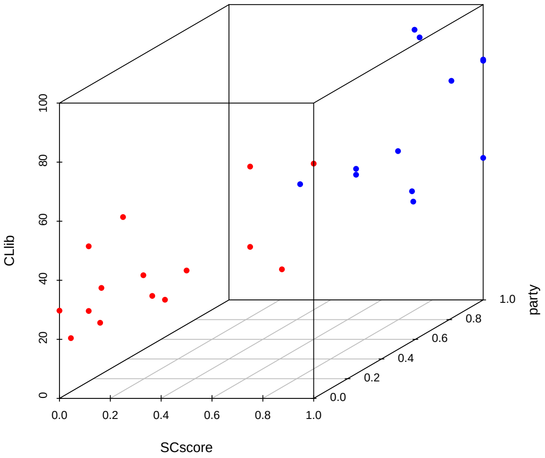
$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i changes with the inclusion of more variables.

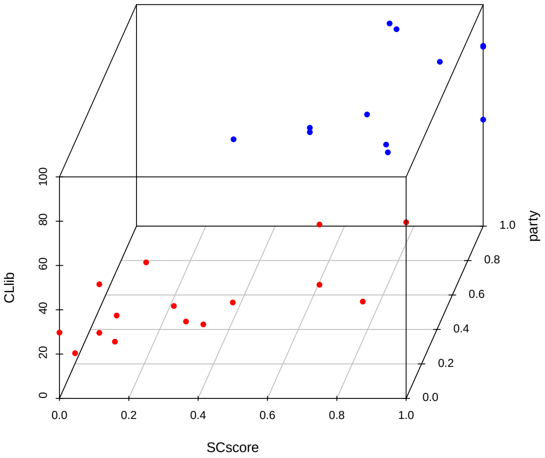
One Binary and One Continuous Conditioning Variable



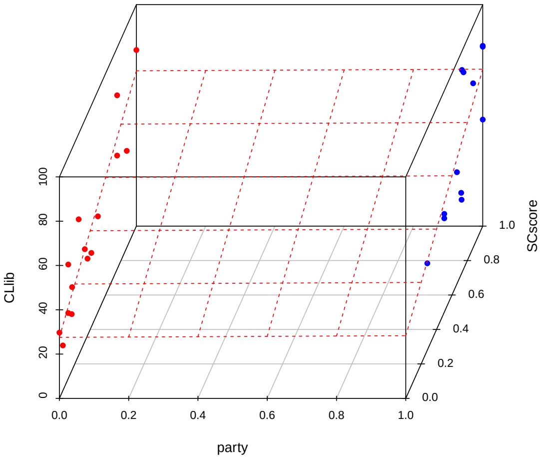
One Binary and One Continuous Conditioning Variable



One Binary and One Continuous Conditioning Variable



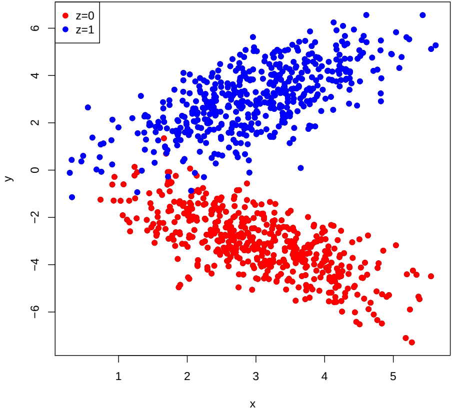
Regression Plane



Misspecification Activity

Draw the two regression lines (i.e., edges of the regression plane) implied by

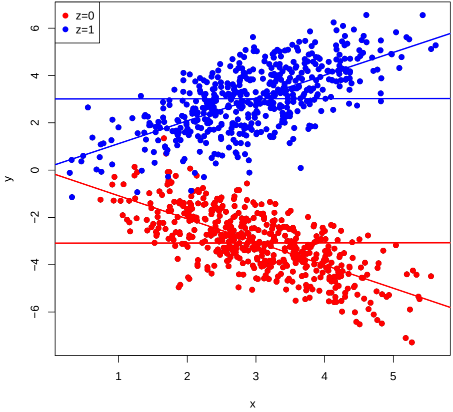
$$\hat{y} = B_0 + B_1 x + B_2 z$$



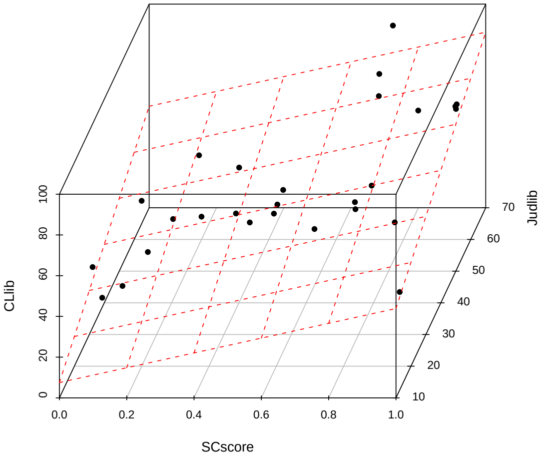
Misspecification Activity

Draw the two regression lines (i.e., edges of the regression plane) implied by

$$\hat{y} = B_0 + B_1 x + B_2 z$$

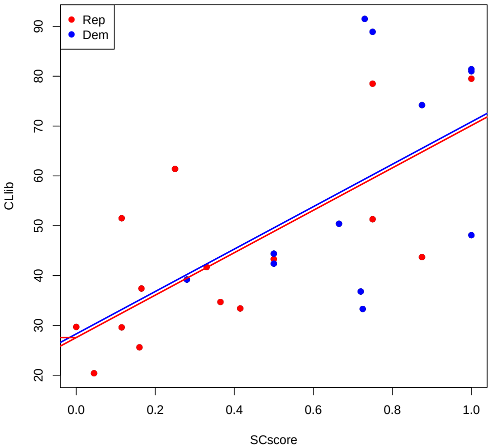


Regression Plane for Two Continuous Conditioning Variables



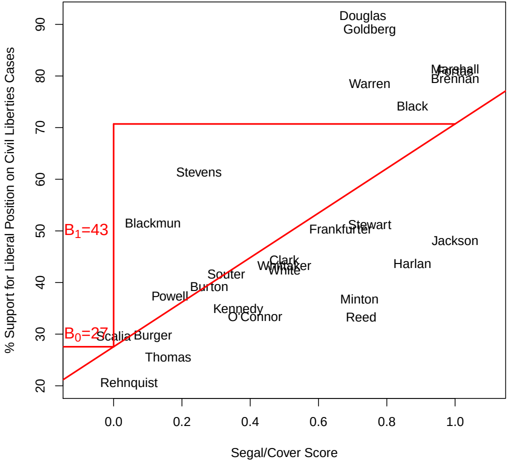
Descriptive Interpretation of Additive Regression

$$\widehat{CLlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



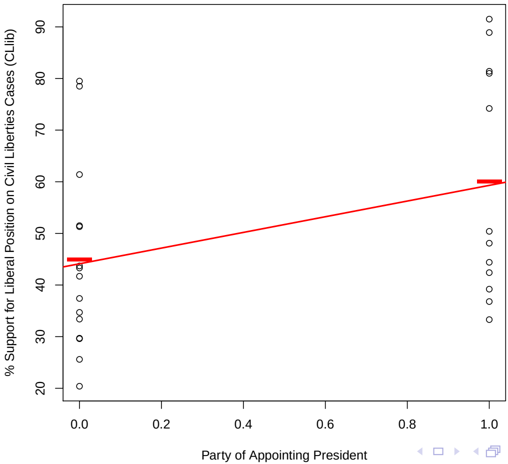
Recall: Simple Regression with SCscore

$$\widehat{CLlib}_i = B_0 + B_1 \cdot SCscore_i$$



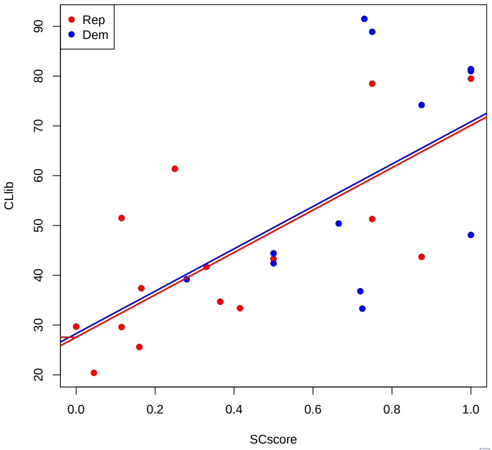
Recall: Simple Regression with Party

$$\widehat{CLlib}_i = B_0 + B_1 \cdot Dem_i$$



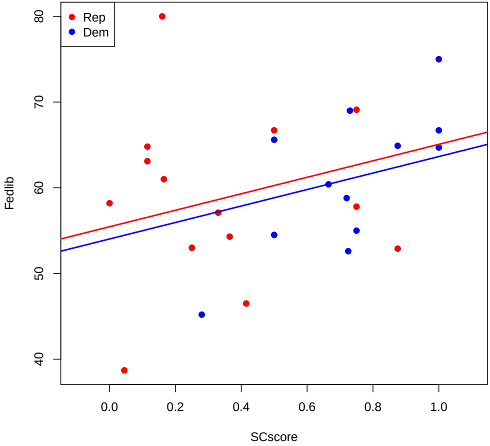
Descriptive Interpretation of Additive Regression

$$\widehat{CLlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



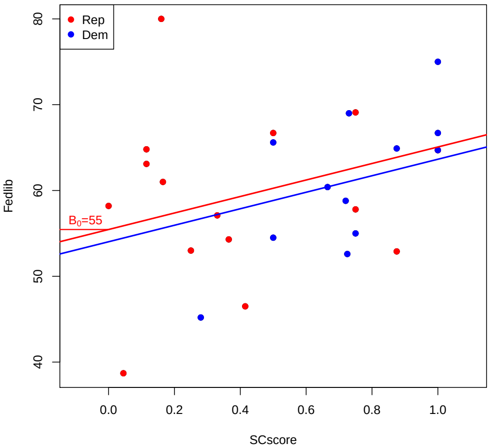
Descriptive Interpretation of Additive Regression

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



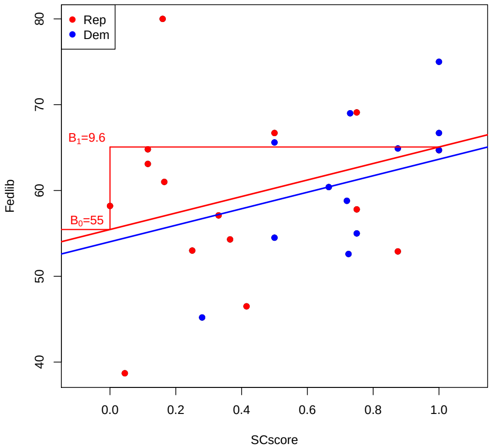
Descriptive Interpretation of Additive Regression

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



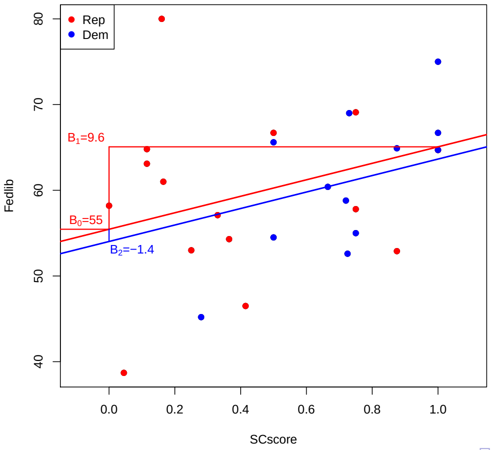
Descriptive Interpretation of Additive Regression

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



Descriptive Interpretation of Additive Regression

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i$$



Additive Interpretations

- B_0: Mean outcome for a Rep justice with a zero SCscore.
- B_1: Mean difference in outcomes between justices appointed by the same party, but with a one unit difference in SCscore.
- B_2: Mean difference in outcomes between justices with the same SCscore, but appointed by different parties (Dem minus Rep).

Discuss the additive linear model assumptions implicit in these interpretations.

Outline

Additive Multiple Linear Regression

Interactive Multiple Linear Regression

Adding/Removing Additive Variables

Multiple Regression with an Interaction

- Fitted Values: $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i + B_3 x_i \cdot z_i$ for $i = 1, ..., N$
- Residuals: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, B_2, and B_3, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i has again changed.

Multiple Regression with an Interaction

- Fitted Values: $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i + B_3 x_i \cdot z_i$ for $i = 1, ..., N$
- Residuals: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, B_2, and B_3, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i has again changed.

Multiple Regression with an Interaction

- Fitted Values: $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i + B_3 x_i \cdot z_i$ for $i = 1, ..., N$
- Residuals: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

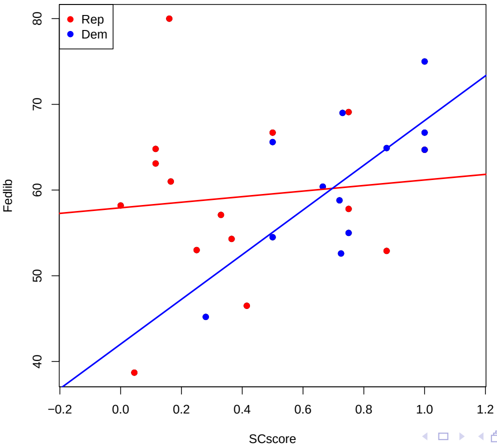
Again, we choose B_0, B_1, B_2, and B_3, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2$$

but the definition of e_i has again changed.

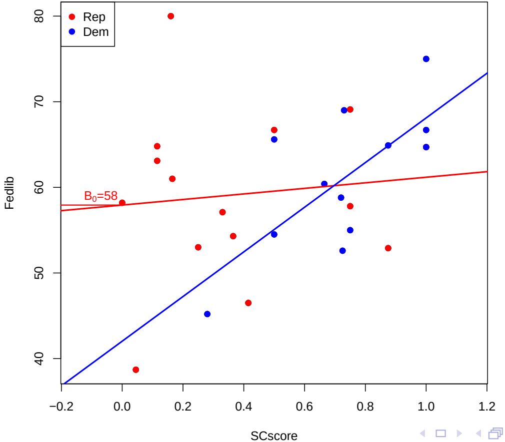
Descriptive Interpretation of Interaction Model

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i + B_3 \cdot SCscore_i \cdot Dem_i$$



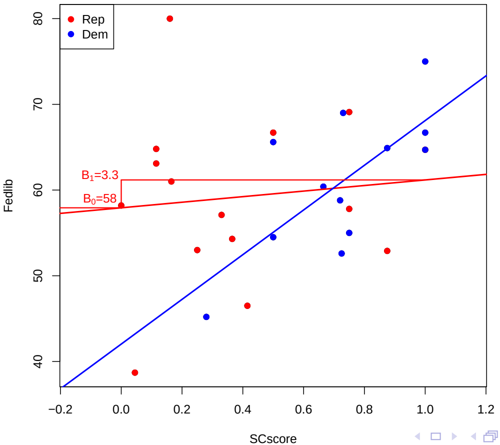
Descriptive Interpretation of Interaction Model

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i + B_3 \cdot SCscore_i \cdot Dem_i$$



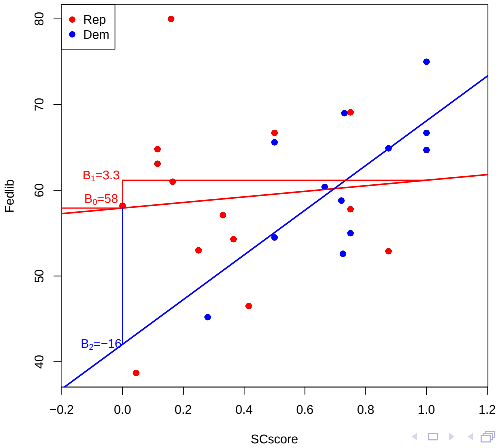
Descriptive Interpretation of Interaction Model

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i + B_3 \cdot SCscore_i \cdot Dem_i$$



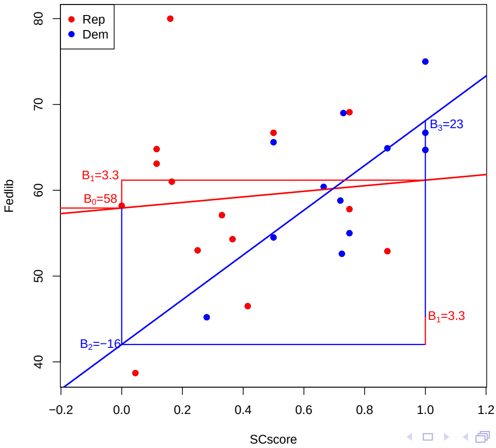
Descriptive Interpretation of Interaction Model

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i + B_3 \cdot SCscore_i \cdot Dem_i$$



Descriptive Interpretation of Interaction Model

$$\widehat{Fedlib}_i = B_0 + B_1 \cdot SCscore_i + B_2 \cdot Dem_i + B_3 \cdot SCscore_i \cdot Dem_i$$



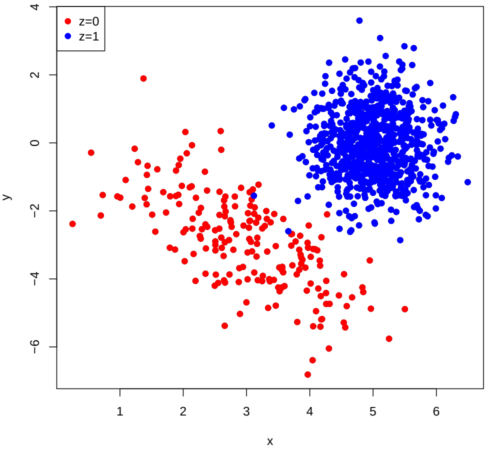
Interactive Interpretations

- B_0: Mean outcome for a Rep justice with a zero SCscore.
- B_1: Mean difference in outcomes between justices appointed by Republicans, but with a one unit difference in SCscore.
- $B_0 + B_2$: Mean outcome for a Dem justice with a zero SCscore.
- B_2: Mean difference in outcomes between justices with a zero SCscore, but appointed by different parties (Dem minus Rep).
- $B_1 + B_3$: Mean difference in outcomes between Dem justices with a one unit difference in SCscore.
- B_3: Mean difference between the mean difference in outcomes between Dem justices with a one unit difference in SCscore and the mean difference in outcomes between Rep justices with a one unit difference in SCscore.

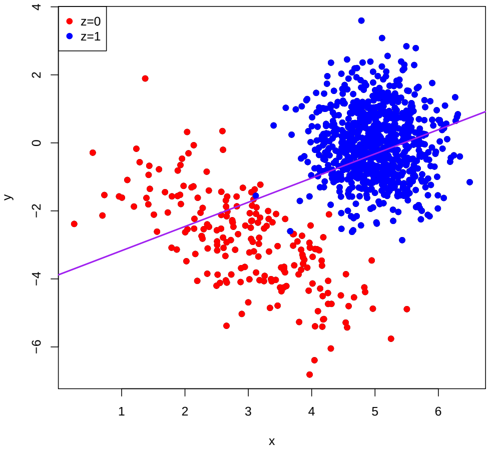
Variance and Multiple Regression

- As before, we want to assess the "leastness" of least squares (i.e., the spread of the data around the plane or twisted plane).
- The standard error of the regression, $\sqrt{\frac{\sum_{i=1}^{N} e_i^2}{N-k}}$, now accounts for the number of B's in the model (where k is the number of B's).
- R^2 uses the same formula $1 - \frac{SSR}{TSS}$, however the inclusion of additional variables will automatically reduce the SSR.

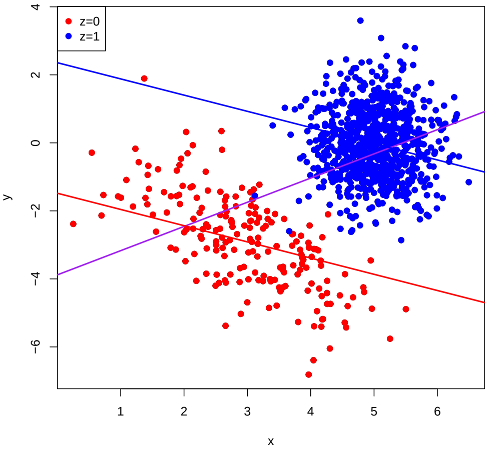
Summary Activity



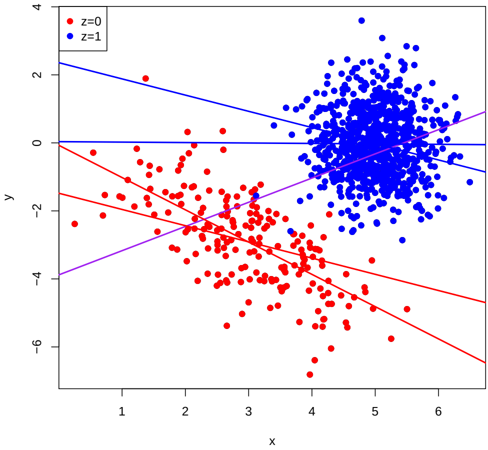
Summary Activity



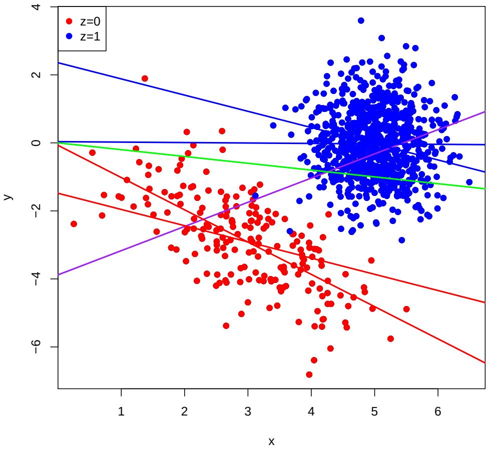
Summary Activity



Summary Activity



Summary Activity



Outline

Multiple Regression with 2 Conditioning Variables

- Fitted Values from the LCMF $\hat{y}_i = B_0 + B_1 x_i + B_2 z_i$ for $i = 1, ..., N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, ..., N$

Again, we choose B_0, B_1, and B_2, to minimize the SSR, where

$$SSR = \sum_{i=1}^{N} e_i^2 = \sum_{i=1}^{N} [y_i - (B_0 + B_1 x_i + B_2 z_i)]^2$$

but the definition of e_i changes with the inclusion of more variables.

Residuals and covariates uncorrelated

$$\frac{\partial SSR}{\partial \tilde{B}_1} = -\sum_{i=1}^{N} 2e_i \cdot x_i$$

$$\frac{\partial SSR}{\partial \tilde{B}_2} = -\sum_{i=1}^{N} 2e_i \cdot z_i$$

We set both equal to zero when solving.

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Then,

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$=$$

$$=$$

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Then,

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)(B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$=$$

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Then,

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)(B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_0}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_1 x_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_2 z_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$+ \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)e_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Then,

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)(B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$= \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_0}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_1 x_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)B_2 z_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

$$+ \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)e_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2} = B_1 + B_2 \cdot \frac{\sum_{i=1}^{n}(x_i - \bar{x}_1)z_i}{\sum_{i=1}^{n}(x_i - \bar{x}_1)^2}$$

The Social Pressure Experiment

Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment

ALAN S. GERBER *Yale University*
DONALD P. GREEN *Yale University*
CHRISTOPHER W. LARIMER *University of Northern Iowa*

Voter turnout theories based on rational self-interested behavior generally fail to predict significant turnout unless they account for the utility that citizens receive from performing their civic duty. We distinguish between two aspects of this type of utility, intrinsic satisfaction from behaving in accordance with a norm and extrinsic incentives to comply, and test the effects of priming intrinsic motives and applying varying degrees of extrinsic pressure. A large-scale field experiment involving several hundred thousand registered voters used a series of mailings to gauge these effects. Substantially higher turnout was observed among those who received mailings promising to publicize their turnout to their household or their neighbors. These findings demonstrate the profound importance of social pressure as an inducement to political participation.

The Civic Duty Treatment

Dear Registered Voter:

DO YOUR CIVIC DUTY AND VOTE!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

The whole point of democracy is that citizens are active participants in government; that we have a voice in government. Your voice starts with your vote. On August 8, remember your rights and responsibilities as a citizen. Remember to vote.

DO YOUR CIVIC DUTY — VOTE!

The Hawthorne Treatment

Dear Registered Voter:

YOU ARE BEING STUDIED!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse.

This year, we're trying to figure out why people do or do not vote. We'll be studying voter turnout in the August 8 primary election.

Our analysis will be based on public records, so you will not be contacted again or disturbed in any way. Anything we learn about your voting or not voting will remain confidential and will not be disclosed to anyone else.

DO YOUR CIVIC DUTY — VOTE!

The Self Treatment

Dear Registered Voter:

WHO VOTES IS PUBLIC INFORMATION!

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse.

This year, we're taking a different approach. We are reminding people that who votes is a matter of public record.

The chart shows your name from the list of registered voters, showing past votes, as well as an empty box which we will fill in to show whether you vote in the August 8 primary election. We intend to mail you an updated chart when we have that information.

We will leave the box blank if you do not vote.

DO YOUR CIVIC DUTY — VOTE!

OAK ST	Aug 04	Nov 04	Aug 06
9999 ROBERT WAYNE		Voted	_______
9999 LAURA WAYNE	Voted	Voted	_______

The Neighbors Treatment

Dear Registered Voter:

WHAT IF YOUR NEIGHBORS KNEW WHETHER YOU VOTED?

Why do so many people fail to vote? We've been talking about the problem for years, but it only seems to get worse. This year, we're taking a new approach. We're sending this mailing to you and your neighbors to publicize who does and does not vote.

The chart shows the names of some of your neighbors, showing which have voted in the past. After the August 8 election, we intend to mail an updated chart. You and your neighbors will all know who voted and who did not.

DO YOUR CIVIC DUTY — VOTE!

MAPLE DR	Aug 04	Nov 04	Aug 06
9995 JOSEPH JAMES SMITH	Voted	Voted	______
9995 JENNIFER KAY SMITH		Voted	______
9997 RICHARD B JACKSON		Voted	______
9999 KATHY MARIE JACKSON		Voted	______
9999 BRIAN JOSEPH JACKSON		Voted	______
9991 JENNIFER KAY THOMPSON		Voted	______
9991 BOB R THOMPSON		Voted	______
9993 BILL S SMITH			______
9989 WILLIAM LUKE CASPER		Voted	______
9989 JENNIFER SUE CASPER		Voted	______
9987 MARIA S JOHNSON	Voted	Voted	______
9987 TOM JACK JOHNSON	Voted	Voted	______
9987 RICHARD TOM JOHNSON		Voted	______

Balance on Pre-treatment Variables in the Gerber, Green, and Larimer Experiment

TABLE 1. Relationship between Treatment Group Assignment and Covariates (Household-Level Data)

	Control	Civic Duty	Hawthorne	Self	Neighbors
	Mean	Mean	Mean	Mean	Mean
Household size	1.91	1.91	1.91	1.91	1.91
Nov 2002	.83	.84	.84	.84	.84
Nov 2000	.87	.87	.87	.86	.87
Aug 2004	.42	.42	.42	.42	.42
Aug 2002	.41	.41	.41	.41	.41
Aug 2000	.26	.27	.26	.26	.26
Female	.50	.50	.50	.50	.50
Age (in years)	51.98	51.85	51.87	51.91	52.01
N =	99,999	20,001	20,002	20,000	20,000

Note: Only registered voters who voted in November 2004 were selected for our sample. Although not included in the table, there were no significant differences between treatment group assignment and covariates measuring race and ethnicity.

Including Covariates in the Gerber, Green, and Larimer Experiment

TABLE 3. OLS Regression Estimates of the Effects of Four Mail Treatments on Voter Turnout in the August 2006 Primary Election

	Model Specifications		
	(a)	(b)	(c)
Civic Duty Treatment (Robust cluster standard errors)	.018* (.003)	.018* (.003)	.018* (.003)
Hawthorne Treatment (Robust cluster standard errors)	.026* (.003)	.026* (.003)	.025* (.003)
Self-Treatment (Robust cluster standard errors)	.049* (.003)	.049* (.003)	.048* (.003)
Neighbors Treatment (Robust cluster standard errors)	.081* (.003)	.082* (.003)	.081* (.003)
N of individuals	344,084	344,084	344,084
Covariates**	No	No	Yes
Block-level fixed effects	No	Yes	Yes

Note: Blocks refer to clusters of neighboring voters within which random assignment occurred. Robust cluster standard errors account for the clustering of individuals within household, which was the unit of random assignment.
* $p < .001$.
** Covariates are dummy variables for voting in general elections in November 2002 and 2000, primary elections in August 2004, 2002, and 2000.

FWL Theorem in Pictures

Results:

$$\hat{\beta}_X^{Y \sim X+Z} = \hat{\beta}_{r_{X\sim Z}}^{Y \sim r_{X\sim Z}}$$

$$\hat{\beta}_X^{Y \sim X+Z} = \hat{\beta}_{r_{X\sim Z}}^{r_{Y\sim Z} \sim r_{X\sim Z}}$$

Simulating Linear Model Data

```
> n <- 10000
> set.seed(1234)
> z <- rnorm(n)
> x <- z + rnorm(n)
> y <- x + z + rnorm(n)
> cor(x,z)
[1] 0.7044271
```

Linear Regression Output

```
> lm(y ~ x)

Coefficients:
(Intercept)            x
    0.01099      1.48301

> lm(y ~ x + z)

Coefficients:
(Intercept)            x            z
    0.00417      0.99901      0.98608
```

Added Variable “Plot” (Partial Regression “Plot”)

```
> r_yz <- residuals(lm(y ~ z))
> r_xz <- residuals(lm(x ~ z))
> r_yxz <- residuals(lm(y ~ x+z))
> lm(r_yz ~ r_xz)

Coefficients:
(Intercept)          r_xz
 -3.619e-16    9.990e-01

> range(r_yxz - residuals(lm(r_yz ~ r_xz)))
[1] -2.211564e-13  4.019007e-14
```

Only x residualized

```
> lm(y ~ r_xz)

Coefficients:
(Intercept)         r_xz
   0.008571     0.999006

> range(r_yxz - residuals(lm(y ~ r_xz)))
[1] -7.209635  6.790846
```

What if x and z are uncorrelated?

What if x and z perfectly correlated?

What if x and z are uncorrelated?

$$\hat{\beta}_X^{Y \sim X} = \hat{\beta}_X^{Y \sim X+Z}$$

Why? (see whiteboard)

What if x and z perfectly correlated?

What if x and z are uncorrelated?

$$\hat{\beta}_X^{Y \sim X} = \hat{\beta}_X^{Y \sim X+Z}$$

Why? (see whiteboard)

What if x and z perfectly correlated?

We can't fit a regression!

What if x and z are uncorrelated?

```
> z <- rnorm(n)
> x <- rnorm(n) # no z
> cor(x,z)
[1] 0.01075281
> y <- x + z + rnorm(n)
> lm(y ~ x)
Coefficients:
(Intercept)            x
    0.01027      1.00940

> lm(y ~ x + z)

Coefficients:
(Intercept)            x            z
    0.00417      0.99901      0.98509
```

Summary

- Additive linear multiple regression is a parsimonious summary of the relationship of multiple variables with an outcome, however care must be taken with the "simple" interpretation.
- Interactive linear multiple regression provides a more flexible summary but complicates the interpretation. However, sometimes we can average over the complications.
- Within additive models, the omitted variable formula and FWL theorem provide insight into the effect of controlling for an additional variable. Stable results can imply research credibility.